



Contents lists available at ScienceDirect

Research in Developmental Disabilities

journal homepage: www.elsevier.com/locate/redevdis

Do maternal and paternal education protect against developmental delays?

Diane L. Putnick^{a,*}, Neil J. Perkins^b, Edwina Yeung^a, Shyamal D. Peddada^b^a Epidemiology Branch, Division of Population Health Research, Division of Intramural Research, Eunice Kennedy Shriver National Institute of Child Health and Human Development, Bethesda, MD, USA^b Biostatistics and Bioinformatics Branch, Division of Population Health Research, Division of Intramural Research, Eunice Kennedy Shriver National Institute of Child Health and Human Development, Bethesda, MD, USA

ARTICLE INFO

Keywords:

Parental education
Developmental delay
Constrained inference

ABSTRACT

Background: Parental education is linked to child development and wellbeing, but unique and combined contributions of maternal and paternal education have rarely been studied.

Aims: This study assessed the individual and joint associations of maternal and paternal education on children's odds of a probable developmental delay.

Methods and procedures: Mothers ($N = 3566$) from the Upstate KIDS cohort study reported on theirs and the father's highest level of education and completed the Ages and Stages Questionnaires up to 7 times when the child was 4- to 36-months-old. A nonparametric constrained inference technique assessed the association between education and probable developmental delays.

Outcomes and results: Higher levels of maternal and paternal education were protective against child probable developmental delays in years 2 (18, 24 months; p -trends $<.001$) and 3 (30, 36 months; p -trends $<.001$), but not year 1 (4, 8, 12 months; p -trend=.486–.832). Trends held adjusting for parental age and race. When adjusting for the other parent's education, only maternal education had unique associations with delays. Among parents with the same or similar education levels, when both parents had one level higher education it was also protective.

Conclusions and implications: Both maternal and paternal education may protect against children's developmental delays in toddlerhood, but maternal education may be particularly important.

What this paper adds?

Maternal education has been linked to wide-ranging aspects of child development. Paternal education's contributions to child development have been studied less, and the combination of maternal and paternal education even less. This paper uses a novel application of multivariate nonparametric constrained inference to assess how maternal and paternal education contribute to children's probable developmental delays alone, controlling for the other parent's education, and in combination. Neither parent's education was associated with developmental delays in the first year, but both parents' educations (alone and jointly) were associated in years 2 and 3, suggesting that early appearing delays may have different contributors than later ones. Only maternal education was uniquely associated with children's developmental delays after adjusting for father education (and not vice versa), suggesting that maternal education may be particularly important for early child development.

* Correspondence to: Epidemiology Branch, Division of Population Health Research, Division of Intramural Research, Eunice Kennedy Shriver National Institute of Child Health and Human Development, 6710B Rockledge Drive, Rm 3137D, Bethesda, MD 301-496-629, USA

E-mail address: putnickd@mail.nih.gov (D.L. Putnick).

<https://doi.org/10.1016/j.ridd.2024.104848>

Received 14 February 2024; Received in revised form 4 September 2024; Accepted 18 September 2024

Available online 26 September 2024

0891-4222/Published by Elsevier Ltd.

1. Introduction

Parental education has been heralded as one of the most influential characteristics for the promotion of child survival, health, and development (Bizzego et al., 2021; Chou et al., 2010; Roberts et al., 1999). What is less clear is whether one or the other parent's education is more important or whether one parent with higher education is sufficient to promote child development. Most research has focused on maternal education (e.g., Zablotsky et al., 2019), and most often education is used only as a covariate in studies of child development. This study explores the associations between maternal and paternal education level with the odds of their child screening positive for a developmental delay in the first 3 years of life.

Maternal education is often used as a proxy for socioeconomic status (SES) because associations of SES with child development are largely driven by maternal education (Bornstein et al., 2003). However, maternal education is a marker of more than just SES. Mothers with the highest levels of education have access to more workplace flexibilities than mothers with less education (Fuller & Hirsh, 2018), which can impact their parenting (Kim, 2020). Mothers with more education make different parenting choices for their children (Cuartas, 2022; Jeong et al., 2017). More educated mothers value and promote behaviors that support their children even before they are born. Mothers with higher levels of education are less likely to smoke during pregnancy (Carneiro et al., 2013), and they eat higher quality foods (Rifas-Shiman et al., 2009) and breastfeed longer (Carneiro et al., 2013). More educated mothers spend more time with their children (Guryan et al., 2008; Kalil et al., 2012). They also enrich the home environment in various ways (Telias et al., 2022), select higher quality childcare (Augustine et al., 2009; Carneiro et al., 2013), ensure earlier bedtimes and routines so children get enough sleep (Hale et al., 2009; McDonald et al., 2014), restrict screen media (Hoyos Cillero & Jago, 2010; Trinh et al., 2020), talk more to their children and use more complex sentences (Rowe, 2012), and read more to their children (Pew Research Center, 2015). Each of these individual behaviors have been shown to promote child development, and the combination of these behaviors in more educated mothers compound to produce relatively large effects on child development (Sonander & Claesson, 1999).

The contributions of paternal education to child development have been much less studied, but evidence also suggests that paternal education contributes to child growth (Jeong et al., 2018; Li et al., 2020; Vollmer et al., 2017), school retention (Chevalier, 2013), language and cognitive development (Edwards & Roff, 2010; Jeong et al., 2018, 2017; Pancsofar et al., 2010), developmental delays (Demirci & Kartal, 2018; Ozkan et al., 2012), and health behaviors (Rammohan et al., 2012). However, paternal education may be less-strongly associated with child development than maternal education (Chevalier, 2013; Jeong et al., 2018) or it may not be associated with child development independent of maternal education (Li et al., 2020; Lundborg et al., 2014). Joint effects of maternal and paternal education remain to be explored.

The age of the child when developmental delays were assessed may be important. Parenting tasks differ across child development. In the first year, much of parenting centers around nurturing, feeding, and setting routines (Bornstein et al., 2022). As children enter the second and third years, parenting gradually adjusts to meet the child's burgeoning skills in motor, language, socioemotional, and cognitive domains (Staples & Bates, 2018). Consequently, the determinants of developmental delays may also vary over development. Early appearing delays may be a consequence of prematurity (Hochstedler et al., 2021), birth weight (Flensburg-Madsen & Mortensen, 2017), and other organic causes (Aicardi, 1998). Delays that appear later or persist into the second and third year may be more responsive to the environments provided by parents – environments that vary with parental education (Carneiro et al., 2013; Zadeh et al., 2010).

Men and women tend to have children with partners who have similar phenotypic characteristics, including education (Eika et al., 2019). Consequently, there is typically a positive correlation coefficient between maternal and paternal education in the range of $r = .40$ – $.60$ (Edwards & Roff, 2010; Pancsofar et al., 2010; Pougnet et al., 2011). Given this overlap, there are several potential hypotheses about how parental education impacts child development. It might be that maternal education is the driving factor because mothers are typically the primary caregivers of young children and spend more time interacting with them (Craig, 2016; Pew Research Center, 2013). Paternal education may be more important because it is a proxy for household resources, which may be decreased by mothers stepping out of jobs to care for young children. Finally, it may be that having any parent (regardless of mother or father) with higher education promotes child development because that parent will apply their knowledge to advocate for the needs of the child. This study will explore the interplay between maternal and paternal education and their impacts on child developmental delays.

In contrast to previous studies, here the ordinal nature of maternal and paternal education will be leveraged. Although some studies have coded education in years allowing it to be modeled continuously (e.g., Edwards & Roff, 2010; Pancsofar et al., 2010; Pougnet et al., 2011), most studies of increasing education levels (e.g., less than high school; high school graduate; college graduate; advanced education) employ dummy coding with a single reference group that does not take advantage of the ordered nature of the data (e.g., Demirci & Kartal, 2018; Ozkan et al., 2012). In this study, we use a multivariate stochastic ordering modeling technique (Davidov & Peddada, 2013a, 2013b) to preserve the ordering of education and test whether increasing levels of maternal and paternal education protect against their children experiencing developmental delays.

2. Material and methods

2.1. Participants

This is a secondary analysis of data from the Upstate KIDS Study, a prospective population-based cohort of children born in

2008–2010, originally designed to evaluate the impact of infertility treatment on child growth and development (Buck Louis et al., 2014). Infants conceived by infertility treatment were frequency matched by birth region and plurality in a 1:3 ratio to those not conceived by infertility treatment. A total of 5034 mothers (27.2 % of 18,479 approached) and 6171 children were recruited into the study when children were 4 months old. A previous paper using the cohort assessed the impact of infertility treatment on children's probable developmental delays and found no associations when accounting for plurality (Yeung et al., 2016), suggesting the full cohort could be combined to explore associations between parental education and probable delays. Due to differences in trajectories of developmental delays in twins (Nan et al., 2013), this secondary analysis was restricted to families with a singleton child with maternal and paternal education data, and probable developmental delay data on the Ages and Stages Questionnaire (see below) available for at least one point in the first, second, and/or third year ($n=3566$).

Of the sample of 3566, 47.8 % of children were girls, about half (54.45 %) were firstborn in their families, and most were born at term (gestational age $M = 38.73$, $SD = 1.88$, range = 23–42 weeks). Most (77.05 %) mothers were married or in a civil union and had private medical insurance (74.47 %), and 26.53 % used infertility treatment to achieve the pregnancy. Additional demographics are presented in Table 1 and discussed below. Human subjects research approval was obtained from all participating institutions (NYSDOH IRB #07-097; UAlbany #08-179), and informed consent was obtained prior to data collection.

2.2. Measures

2.2.1. Parental education

At study entry, mothers self-reported their highest level of education and reported the father's highest level of education. Fathers were defined as "the father who lives with your child (whether or not this is the same person as the child's biological father)." Education was rated on an 8-point scale ranging from 8th grade or less to doctoral or professional (MD, JD) degree. To have conceptually meaningful categories, and due to small counts in certain categories, education was recoded into 5 categories: (1) less than high school, (2) high school graduate or GED, (3) some college or technical school or associate's degree, (4) bachelor's degree, and (5) master's, doctoral, or professional degree. In 104 U.S. mother-father pairs, McKean-Cowdin, Preston-Martin, Pogoda, Mueller, and Holly (2000) found that mothers were reasonable proxy reporters of fathers' education level ($\kappa=.71$).

2.2.2. Child development

Mothers completed the Ages and Stages Questionnaire®, second edition (ASQ-2) when children were 4, 8, and 12 months (Gollenberg, Lynch, Jackson, McGuinness, & Msall, 2010) and third edition (ASQ-3) at 18, 24, 30, and 36 months (Squires & Bricker,

Table 1
Parental education and developmental delay descriptive statistics.

	n (%), unless otherwise noted			
	Overall (<i>N</i> = 3566)	First Year (<i>n</i> = 3532)	Second Year (<i>n</i> = 2107)	Third Year (<i>n</i> = 1878)
<i>Maternal Education</i>				
Less than high school	193 (5.41)	190 (5.38)	64 (3.04)	58 (3.09)
High school diploma or GED	473 (13.26)	470 (13.31)	204 (9.68)	184 (9.80)
Some college, technical school, or associate's degree	1121 (31.44)	1106 (31.31)	589 (27.95)	525 (27.96)
Bachelor's degree	768 (21.54)	761 (21.55)	528 (25.06)	458 (24.39)
Master's, doctoral, or professional degree	1011 (28.35)	1005 (28.45)	722 (34.27)	653 (34.77)
<i>Paternal Education</i>				
Less than high school	318 (8.92)	313 (8.86)	132 (6.26)	109 (5.80)
High school diploma or GED	706 (19.80)	701 (19.85)	342 (16.23)	307 (16.35)
Some college, technical school, or associate's degree	1124 (31.52)	1112 (31.48)	638 (30.28)	579 (30.83)
Bachelor's degree	778 (21.82)	770 (21.80)	535 (25.39)	464 (24.71)
Master's, doctoral, or professional degree	640 (17.95)	636 (18.01)	460 (21.83)	419 (22.31)
<i>Child Developmental Delay in any Domain</i>	781 (21.90) ^a	537 (15.20)	252 (11.96)	191 (10.17)
<i>Covariates</i>				
<i>Mother race/ethnicity</i>				
Non-Hispanic White	2902 (81.38)	2879 (81.51)	1784 (84.67)	1605 (85.46)
Non-Hispanic Black	158 (4.43)	155 (4.39)	60 (2.85)	47 (2.50)
Hispanic, any race	307 (8.61)	302 (8.55)	148 (7.02)	124 (6.60)
Other	199 (5.58)	196 (5.55)	115 (5.46)	102 (5.43)
<i>Father race/ethnicity</i>				
Non-Hispanic White	2825 (79.31)	2801 (79.39)	1762 (83.71)	1578 (84.07)
Non-Hispanic Black	212 (5.95)	209 (5.92)	83 (3.94)	72 (3.84)
Hispanic, any race	339 (9.52)	334 (9.47)	153 (7.27)	134 (7.14)
Other	186 (5.22)	184 (5.22)	107 (5.08)	93 (4.95)
Mother age in years, <i>M</i> (<i>SD</i>)	30.19 (6.03)	31.18 (6.02)	30.94 (5.77)	30.86 (5.81)
Father age in years, <i>M</i> (<i>SD</i>)	32.83 (6.81)	32.82 (6.80)	33.27 (6.54)	33.24 (6.56)

Note. Father race/ethnicity was missing for 4 participants overall and in Year 1, 2 participants in Year 2, and 1 participant in Year 3. Father age was missing for 225 participants overall, 222 in Year 1, 100 in Year 2 and 85 in Year 3.

^a Probable delay at any time in any domain in the first 3 years.

2009). Preterm children were assessed at term-corrected ages. The ASQ-3 was released mid-way through data collection and the two ASQ versions were very similar for the age groups we studied, hence changing versions is unlikely to affect the results. The ASQ is a validated screening instrument designed to detect developmental impairments in five domains: fine motor, gross motor, communication, personal-social functioning, and problem-solving skills (Yue et al., 2019). ASQ items were scored as “yes” (10 points), “sometimes” (5 points), and “not yet” (0 points). On a given domain of the ASQ, a probable developmental delay is defined by a score that is two or more standard deviations below the United States national average for that development area and the specified age group. Dichotomous scores were used rather than continuous scores because the questions and cut-off values for probable delay vary across child age, suggesting that the total score does not have the same meaning over time (measurement non-invariance). For example, the cutoff score for probable delay on the gross motor domain at 12 months is 18, at 24 months is 38, and at 36 months is 37 (all out of 60 possible points). These widely varying cutoffs (all representing 2 standard deviations below the national mean) suggest that the total scores are not equivalent across age and should not be used to indicate developmental performance.

A total ASQ delay score was computed as having a probable developmental delay on any domain of the ASQ. Due to potential differences in parental education's influence across child age, probable developmental delays in years 1, 2 and 3 were coded into age ranges. Low ASQ scores at 4, 8, and/or 12 months were coded as having a probable delay in year 1; low scores at 18 and/or 24 months were coded as having a probable delay in year 2; low scores at 30 and/or 36 months were coded as having a probable delay in year 3.

2.2.3. Covariates

A baseline questionnaire at 4 months assessed demographic factors, and vital records were abstracted. Mother and father ages were primarily obtained from vital records and supplemented with maternal reports when necessary. Mothers reported theirs and the father's race/ethnicity, which were dichotomized into non-Hispanic white versus other due to small numbers in other groups (Table 1). Vital records supplemented these maternal reports as available and needed.

2.3. Analytic plan

As shown in Table 1, there was a great deal of attrition in the study with nearly half (47 %) of the data missing by the 3rd year. Because families with lower education were more likely to be missing in the 2nd and 3rd years, χ^2 s (4, $n=3566$) = 124.87–216.57, p s < .001, it was important to account for missing data. Therefore, multiple imputation by chained equations was utilized to generate 10 datasets using the mice package version 3.14.0 (van Buuren & Groothuis-Oudshoorn, 2011) in R version 4.2.1. Results from the 10 imputed datasets were combined using Rubin's rules (Rubin, 1987). As a sensitivity analysis, we ran the analyses on data that was multiply imputed only for parental education and covariates and the results were stronger (more significant; see Tables S1 and S2 and Figure S1). We concluded that imputing the outcomes as well as parental education and covariates was the more conservative approach and present these results in the main paper.

A priori, higher education levels are assumed to be protective against developmental delay or at the very least not harmful. This assumption is based on the extensive literature about the benefits of increasing maternal education for child development, even at its extremes (Basu & Stephenson, 2005). Thus, we tested one-sided hypotheses in the protective direction versus no effect rather than the traditional two-sided hypotheses. Traditional approaches test one hypothesis at a time, perform two-sided tests, and interpret the directions for the parameters based on the sign of the estimated parameters. As demonstrated in Davidov and Peddada (2011, 2013a, 2013b) such approaches are not only inefficient in terms of loss of power at the expense of potentially inflated false positive rate, but not always convenient to interpret the data. For these reasons, we take a nonparametric multivariate approach to the problem where all binary contrasts are tested together, and in each case a directional hypothesis is tested according to the question of interest. In doing so, we not only control the overall false positive rate, but increase the power of the test.

Specifically, to answer the questions of whether the odds of a child's developmental delay decreased as the levels of maternal and paternal education increased, we adopt and extend the methodology developed for multivariate stochastic ordering in Davidov and Peddada (2011, 2013a, 2013b) and Davidov, Jelsema, and Peddada (2018) to adjust for covariates. In this multivariate nonparametric constrained inference technique, for each binary outcome variable we perform standard logistic regression analysis with dummy variables for ordered education levels without and with covariates for unadjusted and adjusted analyses, respectively. The resulting coefficients were then used to estimate contrasts of interest, namely one level increases in education. A vector of these pairwise comparisons represents the unconstrained estimator of the contrasts. Constrained estimation desired here is accomplished by projecting the vector of unconstrained contrasts onto the set representing the monotonic trend described in the alternative hypothesis using convex programming. This is because the monotonic trend described in alternative hypothesis is a convex polyhedral cone and the idea is to identify and make minimal changes to the unconstrained estimator so that the result vector satisfies the monotonic trend required by the alternative hypothesis. This resulting estimator is called the constrained estimator. A Wald type statistic is constructed using the constrained estimator. Since this test statistic uses the constrained estimators under the alternative hypothesis it has larger power than the standard tests that use the unconstrained estimator. Since the unconstrained estimator is asymptotically multivariate normally distributed, we adopt the methodology developed in Davidov et al. (2018) and Peddada, Dinse, and Haseman (2005) to simulate the asymptotic null distribution of the constrained test to derive p -values for individual and sets of contrasts for a single year or across multiple years. For theoretical details regarding the methodologies used in this paper, we refer the reader to the above papers referenced.

Analyses were performed for children's developmental delays in any domain in year 1, 2, and 3 together and then separately, starting broadly and then drilling down looking for more granular effects across development. First, the associations of maternal and paternal education on developmental delays were estimated in separate models without and with adjustment for parental age, race/

ethnicity, and the education of the other parent. These models were repeated for developmental delays in each individual domain (fine motor, gross motor, communication, personal-social, and problem solving; see Table S3) to determine whether results were similar across domains. Next, we estimated the effects of household education by assessing the joint effects of mother and father education levels. Maternal and paternal education are highly correlated, so we focused on households with the “same” or “similar” education levels (see shaded cells in Table 2). Children of parents with the same levels of education were compared to those who both had one level higher education (e.g., dyads where both parents had less than high school education (level 1) were compared to dyads where both parents were high school graduates (level 2) etc.). A second household-level analysis included children of parents with similar education levels compared to those who both had one level higher education. This was inclusive of the same education level contrasts and added contrasts such as households with maternal and paternal levels 1 and 2, respectively, compared to households with maternal and paternal levels 2 and 3, respectively.

3. Results

Table 1 shows the number and percentage of mothers and fathers who fell into each level of education. Mothers with Master’s, doctoral, or professional degrees were somewhat over-represented in the sample, but each level contained enough mothers and fathers to analyze. When mothers and fathers were cross-tabulated (Table 2), it was evident that most dyads (84.8 %) had levels of education within one level of one-another, with 43.9 % having the same reported education level. Fifteen percent of children were identified as having a probable developmental delay in any domain (fine or gross motor, communication, personal-social, or problem solving) in the first year of life; 12 % in the second year of life, and 10 % in the 3rd year.

3.1. Maternal effects

Protective trends of the effect of maternal education on probable developmental delay were found looking at contrasts across years 1–3, in unadjusted models and models adjusted for maternal age and race/ethnicity and further adjusted for paternal education level, all three models p for trend $< .001$ (first row of Table 3). Associations of maternal education with child probable developmental delays for individual year models as well as individual education contrasts are presented in Table 3. In year 1, there was no evidence of protective associations of maternal education with probable developmental delay in the unadjusted model, p for trend = 0.483. In the adjusted analyses, the model significance was driven by a single contrast – compared to high school graduates, mothers with some college, technical school, or associate’s degree had children with lower odds of probable developmental delay when controlling for maternal age, non-Hispanic White race/ethnicity, and father education. In years 2 and 3, the unadjusted and adjusted models were all significant overall and individual level contrasts were all protective (constrained ORs < 1), but not all contrasts reached significance. In year 2, the contrast between mothers with less than a high school education and mothers who graduated high school and the contrast between mothers with some college, technical school, or associate’s degree and those with a bachelor’s degree were significant. In year 3, increasing levels of education between high school graduation and college graduation were significant, but the contrast between some college, technical school, or associate’s degree and bachelor’s degree attenuated when adjusting for paternal education. In domain-specific models (Tables S4–S8), the overall pattern of effects was similar for fine and gross motor, communication, personal-social, and problem solving delays. Fig. 1A displays the fully adjusted results with a single reference group of high school graduation (level 2) to further illustrate the trends for increasing levels of education.

3.2. Paternal effects

Protective trends of the effect of paternal education on probable developmental delay were found looking at contrasts across years 1–3 (first row in Table 4), in unadjusted models and when adjusting for paternal age and race/ethnicity, p for trend < 0.001 and 0.002 , respectively. However, further adjusting for maternal education level attenuated the effects, p for trend = 0.443. Associations of

Table 2

Cross-tabulation of mother and father education (N = 3566).

Maternal Education	Paternal Education					Total
	1	2	3	4	5	
1. Less than high school	113	58	17	4	1	193
2. High school diploma or GED	96	228	134	13	2	473
3. Some college, technical school, or associate’s degree	97	298	519	165	42	1121
4. Bachelor’s degree	6	70	222	290	180	768
5. Master’s, doctoral, or professional degree	6	52	232	306	415	1011
Total	318	706	1124	778	640	3566

Note. In analyses that included both mothers and fathers, levels 4 and 5 were combined. Darker shaded cells on the diagonal had the same level of education and lighter shaded cells were within one level of education.

Table 3

Constrained odds ratio (OR) estimates ¹ from unadjusted and adjusted models of associations of maternal education with probable developmental delays by child age.

Mother education contrasts	Unadjusted Analysis		Adjusted analysis			
	Constrained OR	p	Constrained OR ²	p	Constrained OR ³	p
ASQ Years 1–3 ⁴		< 0.001*		< 0.001*		< 0.001*
ASQ Year 1 ⁵		0.486		0.007*		0.005*
Level 1 vs. 2	1.00	1.000	1.00	1.000	1.00	0.445
Level 2 vs. 3	0.90	0.106	0.66	<0.001*	0.62	<0.001*
Level 3 vs. 4	1.00	1.000	1.00	0.337	1.00	0.379
Level 4 vs. 5	1.00	0.371	1.00	0.392	0.94	0.223
ASQ Year 2 ⁵		<0.001*		<0.001*		<0.001*
Level 1 vs. 2	0.50	<0.001*	0.49	<0.001*	0.46	0.001*
Level 2 vs. 3	0.91	0.139	0.87	0.123	0.90	0.216
Level 3 vs. 4	0.70	0.001*	0.67	0.001*	0.66	0.005*
Level 4 vs. 5	0.90	0.155	0.89	0.158	0.90	0.208
ASQ Year 3 ⁵		<0.001*		0.002*		0.044*
Level 1 vs. 2	0.73	0.066	0.76	0.116	0.82	0.189
Level 2 vs. 3	0.70	0.005*	0.71	0.009*	0.76	0.040*
Level 3 vs. 4	0.78	0.020*	0.78	0.032*	0.81	0.080
Level 4 vs. 5	0.98	0.319	0.99	0.366	1.00	0.414

1. Estimates are constrained to be one-sided (protective) or no effect.

2. Adjusted models control for maternal age, and non-Hispanic White race/ethnicity vs. other.

3. Adjusted models control for maternal age, non-Hispanic White race/ethnicity vs. other, and paternal education.

4. Trends across all years 1, 2, and 3.

5. Model p-values for monotonic trend in differences across education levels (i.e., as defined in Table 2: level 1 as < high school... and level 5 as Master's or beyond).

* Significant result.

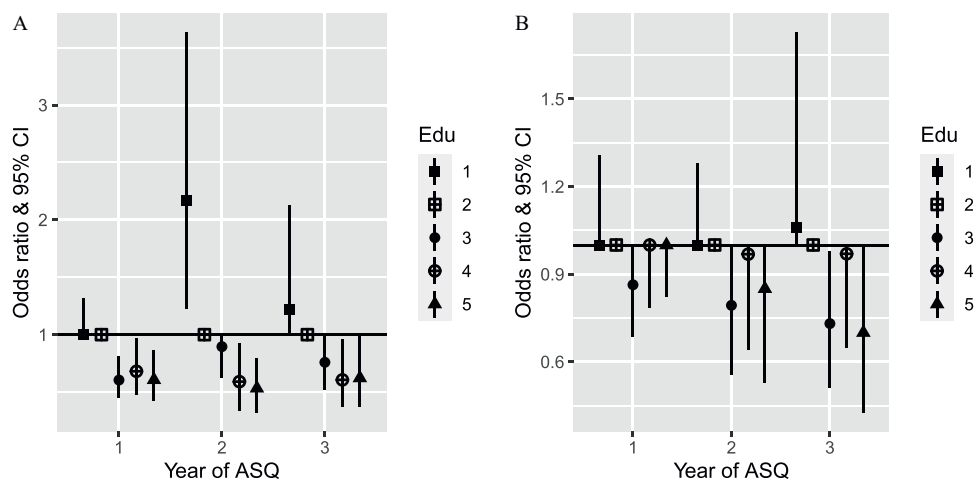


Fig. 1. Unique constrained associations of maternal and paternal education with the odds of probable developmental delay by year. *Note.* Panel A displays the constrained odds of probable developmental delay for levels of maternal education, compared to high school graduation (level 2), adjusting for paternal education and covariates. Panel B displays the constrained odds of probable developmental delay for levels of paternal education, compared to high school graduation (level 2), adjusting for maternal education and covariates. Confidence intervals in this Figure do not exactly match the significance levels in Tables 3 and 4 because the mechanism of generating the null distribution for testing a one-sided hypothesis is not same as the mechanism of generating the distribution of the constrained estimator for constructing confidence intervals.

paternal education with child probable developmental delays for individual years as well as individual education contrasts are presented in Table 4. Higher levels of paternal education were not associated with a lower risk of a probable developmental delay in year 1, p for trend = 0.832, 0.759, and 0.880 in unadjusted and both adjusted models, respectively. However, higher levels of paternal education were associated with a lower risk of a probable developmental delay in unadjusted models in year 2, p for trend < 0.001, and year 3, p for trend < 0.001, as well as adjusting for paternal age and race/ethnicity, both p for trend = 0.002. Additional adjustment for maternal education attenuated the effects, p for trend = 0.440 and 0.134, respectively, for year 2 and 3. In year 2, individual contrasts were only significant for having some college, technical school, or associate's degree versus a high school diploma or GED. In year 3, individual contrasts were significant for having some college, technical school, or associate's degree versus a high school diploma or

Table 4

Constrained odds ratio (OR) estimates ¹ from unadjusted and adjusted models of associations of paternal education with probable developmental delays by child age.

Father education contrasts	Unadjusted Analysis		Adjusted analysis			
	Constrained OR	<i>p</i>	Constrained OR ²	<i>p</i>	Constrained OR ³	<i>p</i>
ASQ Years 1–3 ⁴		< 0.001*		0.002*		0.443
ASQ Year 1 ⁵		0.832		0.759		0.880
Level 1 vs. 2	1.00	0.396	1.00	0.411	1.00	0.440
Level 2 vs. 3	1.00	0.323	0.97	0.252	1.00	0.390
Level 3 vs. 4	1.00	1.000	1.00	0.326	1.00	0.374
Level 4 vs. 5	1.00	0.380	1.00	0.391	1.00	0.419
ASQ Year 2 ⁵		< 0.001*		0.002*		0.440
Level 1 vs. 2	0.97	0.337	1.00	0.430	1.00	1.000
Level 2 vs. 3	0.67	< 0.001*	0.68	< 0.001*	0.85	0.112
Level 3 vs. 4	1.00	0.305	1.00	0.332	1.00	0.388
Level 4 vs. 5	0.82	0.085	0.82	0.088	0.93	0.301
ASQ Year 3 ⁵		< 0.001*		0.002*		0.134
Level 1 vs. 2	0.76	0.063	0.81	0.134	0.90	0.284
Level 2 vs. 3	0.68	0.002*	0.69	0.004*	0.82	0.072
Level 3 vs. 4	1.00	0.338	1.00	0.357	1.00	0.394
Level 4 vs. 5	0.78	0.045*	0.77	0.048*	0.79	0.097

1.Estimates are constrained to be one-sided (protective) or no effect.

2.Adjusted models control for paternal age, and non-Hispanic White race/ethnicity vs. other.

3.Adjusted models control for paternal age, non-Hispanic White race/ethnicity vs. other, and maternal education.

4.Trends across all years 1, 2, and 3.

6.Model *p*-values for monotonic trend in differences across education levels (i.e., as defined in Table 2: level 1 as < high school... and level 5 as Master's or beyond).

* Significant result.

GED and have an advanced degree versus a bachelor's degree. None of the individual contrasts remained significant after adjustment for maternal education. In domain-specific models (Tables S9-S13), the overall pattern of effects was similar for fine and gross motor, communication, and problem solving delays. Unlike in the any delay model, father education remained protective against probable delays in personal-social skills in Year 2 when adjusting for maternal education. Fig. 1B displays the adjusted results with a single reference group of high school graduation (level 2) to further illustrate the absence of trends for increasing levels of education.

3.3. Household effects

To answer the question of whether household, or joint maternal and paternal, education contribute to child probable developmental delays, we explored trends of protective effects of increasing education in households where both parents had the same levels and similar levels of education (Table 5). These analyses for households of same and similar education reduce the overall sample sizes to *n* (%) = 1565 (44 %) and 3024 (85 %), respectively. Overall tests for trends of protective effects of increasing household education on developmental delay across all three years adjusted for maternal and paternal age and race/ethnicity were significant, *p* for trend = 0.025 and < 0.001, respectively for same and similar education level (first row in Table 5).

In individual year models (Table 5), trends failed to reach significance in year 1, *p* for trend = 0.060 and 0.052, for same and similar education levels. In years 2 and 3, trends were significant. These findings suggest that higher levels of parental education (considering mothers and fathers) are associated with a lower risk of probable developmental delay when children are in their second and third years.

Table 5

Test for monotonic trends of protective associations¹ between household education (maternal and paternal) and probable developmental delays from adjusted models² by child age.

	Same Education level	Similar Education level
	<i>p</i>	<i>p</i>
ASQ Years 1–3 ³	0.025	< 0.001
ASQ Year 1 ⁴	0.060	0.052
ASQ Year 2 ⁴	0.016	< 0.001
ASQ Year 3 ⁴	0.047	0.004

1.Estimates are constrained to be one-sided (protective) or no effect.

2.Adjusted models control for maternal and paternal ages, and non-Hispanic White race/ethnicity vs. other.

3.Trends across all years 1, 2, and 3.

4.Trends within individual years.

4. Discussion

This study employed a constrained estimation technique to explore the effects of maternal and paternal education on child probable developmental delays. This technique allowed us to model the odds that increasing levels of education would reduce the likelihood of child developmental delays in children under 3 years. When aggregating across the three years, there were significant joint and individual effects of maternal and paternal education on probable developmental delays when adjusting for age and race/ethnicity, but the individual paternal education effect attenuated when adjusting for maternal education.

When exploring the results by year, we found smaller and less consistent effects of maternal or paternal education on children's probable developmental delays in the first year. This finding is similar to [Slykerman et al. \(2007\)](#) and [Valla, Wentzel-Larsen, Hofoss, and Slinning \(2015\)](#), who both found that maternal education did not predict developmental delays in the first year. Early-appearing delays may have different etiology. For example, links between prematurity ([Valla et al., 2015](#)) and maternal smoking ([Slykerman et al., 2007](#)) with developmental delays in the first year suggest that in-utero exposures or other prenatal risk factors may be at play. Studies have found that up to 40 % of developmental delays have a genetic basis ([Miclea et al., 2015](#); [Rauch et al., 2006](#)) which may also lead to earlier identification.

Unlike in year 1, both maternal and paternal education protected against probable developmental delays in years 2 and 3. In the second and third years of a child's life, parents are setting important foundations for their child's development. For example, parents with higher education talk to their children more ([Magnuson et al., 2009](#)), which in turn promotes child communication and social skills ([Bornstein et al., 1998](#); [Tamis-LeMonda et al., 1996](#)). Parents with higher education also enrich the home environment more, which leads to higher cognitive development ([Telias et al., 2022](#)).

Maternal education effects were uniquely protective against child developmental delays when controlling for paternal education, but paternal education did not have unique associations with probable developmental delays controlling for maternal education. Maternal education may have unique protective effects on children's developmental delays due to mothers' prominence in young children's lives. Mothers are vastly more likely than fathers to be stay-at-home parents ([Cohn et al., 2014](#); [Pew Research Center, 2015](#)), and mothers engage in more childrearing activities than fathers ([Pakaluk & Price, 2020](#); [Pew Research Center, 2023](#)). Studies from China ([Cui et al., 2019](#)) and Sweden ([Lundborg et al., 2014](#)) used compulsory education reforms to estimate the impact of increased maternal education on child health and development. Both studies found that increases in maternal education impacted their children's development (only explored for boys in Sweden). Interestingly, the Swedish study observed similar increases in education for mothers and fathers, but fathers' increases had little effect on their children. The striking differences between the effects of mother and father education in the Swedish study may have been (at least partially) due to the timing of the study. Children were born in the 1960s and 1970s when maternal workforce participation was lower, which may have widened differences between mothers and fathers.

Recognizing that maternal and paternal education are often highly correlated and may work together, we explored the joint effects of maternal and paternal education. Modeling one-level increases in education for both mothers and fathers with the same level and similar (within one-level of one another) education showed that both maternal and paternal education were protective against probable developmental delays in years 2 and 3, even when adjusting for maternal and paternal age and non-Hispanic White race/ethnicity.

4.1. Strengths and limitations

This study was supported by a relatively large sample from across New York state and used a constrained estimation approach to conserve power. However, several methodological limitations were also evident. There was differential attrition over time in that parents with lower education were more likely to drop out of the study. We tried to account for this by using multiple imputation to avoid potential bias. Mothers reported on their child's probable developmental delays, which is common practice, but may have inflated associations between maternal education and children's delays. Father education was also reported by mothers, which may have introduced some error, although mothers are reasonable reporters of fathers' education ([McKean-Cowdin et al., 2000](#)). Furthermore, due to small cell sizes, some education levels had to be combined. Still, we were able to retain meaningful and commonly reported education levels in high-income countries. Generalizability of study findings to other countries, and particularly low- and middle-income countries (LMIC) where educational attainment is typically lower or the disparity between maternal and paternal education is higher, is unclear. Future studies could leverage data from other countries to test whether the patterns found here hold. Parental education has been found to be a powerful predictor of child development in LMIC ([Bizzego et al., 2021](#); [Jeong et al., 2018](#)). Given that mothers engage in more caregiving behaviors than fathers with children under 5 in LMIC ([Bornstein & Putnick, 2016](#)), we might expect to see a stronger associations between maternal than paternal education and early child development in LMIC, and one study across LMIC supports this pattern ([Jeong et al., 2018](#)). Finally, due to the constrained estimation approach used in the study, we focus on *p*-values for one-sided hypothesis tests rather than confidence intervals. The tests employed here implicitly account for margin of error, however, their nature prevents us from providing the traditionally paired standard errors and confidence intervals. The derivation of accurate standard errors and the distribution of the pivotal statistic are necessary for deriving the margin of errors. Since the mechanism of generating the null distribution for testing one-sided hypothesis is not same as the mechanism of generating the distribution of the constrained estimator for constructing confidence intervals, statistical significance from the two approaches may not be identical. This lack of duality is also true when performing statistical hypothesis testing using *F*-test in ANOVA and the statistical tests based on simultaneous confidence intervals for contrasts between the means.

5. Conclusions

Overall, this study suggests that both maternal and paternal education protect against children's probable developmental delays in their children's second and third year, but maternal education may have a particularly important role to play. Understanding that lower parental education, and in particular maternal education, is a risk factor for developmental delays in the second and third year (but not the first year) may help clinicians to decide when and how to monitor and intervene with young children. Evidence suggests that various environmental (Telias et al., 2022; Zadeh et al., 2010) and parenting (Dubow et al., 2009; Harding et al., 2015) factors may mediate the effects of maternal education on child development, and future research should also explore factors that mediate paternal education's effects (Vollmer et al., 2017). Finally, future research should include older children. As children age, fathers may play a more central role in fostering academics, physical development, and socioemotional skills (Beets & Foley, 2008; Flouri & Buchanan, 2004; Leidy, 2013) and their education may become more uniquely influential.

Data Statement

The data necessary to reproduce the analyses presented here are not publicly accessible due to limitations of the consent form, but will be made available upon reasonable request.

Funding

This work was supported by the Intramural Research Program of the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD; contracts #HHSN275201200005C, #HHSN267200700019C).

CRediT authorship contribution statement

Neil J. Perkins: Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Diane L. Putnick:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Shyamal D. Peddada:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Edwina Yeung:** Writing – review & editing, Supervision, Conceptualization.

Data availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ridd.2024.104848](https://doi.org/10.1016/j.ridd.2024.104848).

References

- Aicardi, J. (1998). The etiology of developmental delay. *Seminars in Pediatric Neurology*, 5(1), 15–20. [https://doi.org/10.1016/S1071-9091\(98\)80013-2](https://doi.org/10.1016/S1071-9091(98)80013-2)
- Augustine, J. M., Cavanagh, S. E., & Crosnoe, R. (2009). Maternal education, early child care and the reproduction of advantage. *Social Forces*, 88(1), 1–30. <https://doi.org/10.1353/sof.0.0233>
- Basu, A. M., & Stephenson, R. (2005). Low levels of maternal education and the proximate determinants of childhood mortality: A little learning is not a dangerous thing. *Social Science and Medicine*, 60(9), 2011–2023. <https://doi.org/10.1016/j.socscimed.2004.08.057>
- Beets, M. E., & Foley, J. T. (2008). Association of father involvement and neighborhood quality with kindergartners' physical activity: A multilevel structural equation model. *American Journal of Health Promotion*, 22(3), 195–203. <https://doi.org/10.4278/ajhp.22.3.195>
- Bizzego, A., Gabrieli, G., Bornstein, M. H., Deater-Deckard, K., Lansford, J. E., Bradley, R. H., Costa, M., & Esposito, G. (2021). Predictors of contemporary under-5 child mortality in low- and middle-income countries: A machine learning approach. *International Journal of Environmental Research and Public Health*, 18(3), 1315. <https://doi.org/10.3390/ijerph18031315>
- Bornstein, M. H., Hahn, C.-S., Suwalsky, J. T. D., & Haynes, O. M. (2003). Socioeconomic status, parenting, and child development: The Hollingshead Four-Factor Index of Social Status and The Socioeconomic Index of Occupations. In M. H. Bornstein, & R. H. Bradley (Eds.), *Socioeconomic Status, Parenting, and Child Development* (pp. 29–82). Erlbaum.
- Bornstein, M. H., Haynes, O. M., & Painter, K. M. (1998). Sources of child vocabulary competence: A multivariate model. *Journal of Child Language*, 25(2), 367–393. <https://doi.org/10.1017/s0305000998003456>
- Bornstein, M. H., Haynes, O. M., Park, Y., & Suwalsky, J. T. D. (2022). The United States: Metropolitan and appalachian parenting and infancy. In M. H. Bornstein (Ed.), *Parenting, infancy, culture: Specificity and commonality in Argentina, Belgium, Israel, Italy: and the United States*. Routledge.
- Bornstein, M. H., & Putnick, D. L. (2016). Mothers' and fathers' parenting practices with their daughters and sons in low- and middle-income countries (Mar) *Monographs of the Society for Research in Child Development*, 81(1), 60–77. <https://doi.org/10.1111/mono.12226>
- Buck Louis, G. M., Hediger, M. L., Bell, E. M., Kus, C. A., Sundaram, R., ... Druschel, C. M. (2014). Methodology for establishing a population-based birth cohort focusing on couple fertility and children's development, the Upstate KIDS Study. *Paediatric and Perinatal Epidemiology*, 28(3), 191–202. <https://doi.org/10.1111/ppe.12121>
- Carneiro, P., Meghir, C., & Parey, M. (2013). Maternal education, home environments, and the development of children and adolescents. *Journal of the European Economic Association*, 11, 123–160. <https://doi.org/10.1111/j.1542-4774.2012.01096.x>
- Chevalier, A. (2013). The impact of parental income and education on the schooling of their children. *IZA Journal of Labor Economics*, 2, 8. <https://doi.org/10.1186/2193-8997-2-8>

- Chou, S.-Y., Liu, J.-T., Grossman, M., & Joyce, T. (2010). Parental education and child health: Evidence from a natural experiment in Taiwan. *American Economic Journal: Applied Economics*, 2(1), 33–61. <https://doi.org/10.1257/app.2.1.33>
- Cohn, D. V., Livingston, G., & Wang, W. (2014). Pew Research Center. *After decades of Decline, a rise in Stay-at-home mothers*.
- Craig, L. (2016). Does father care mean fathers share? A comparison of how mothers and fathers in intact families spend time with children. *Gender Society*, 20(2), 259–281. <https://doi.org/10.1177/0891243205285212>
- Cuatas, J. (2022). The effect of maternal education on parenting and early childhood development: An instrumental variables approach. *Journal of Family Psychology*, 36(2), 280–290. <https://doi.org/10.1037/fam0000886>
- Cui, Y., Liu, H., & Zhao, L. (2019). Mother's education and child development: Evidence from the compulsory school reform in China. *Journal of Comparative Economics*, 47(3), 669–692. <https://doi.org/10.1016/j.jce.2019.04.001>
- Davidov, O., Jelsema, C. M., & Peddada, S. (2018). Testing for inequality constraints in singular models by trimming or winsorizing the variance matrix. *Journal of the American Statistical Association*, 113(522), 906–918. <https://doi.org/10.1080/01621459.2017.1301258>
- Davidov, O., & Peddada, S. (2011). Order restricted inference for multivariate binary data with application to toxicology. *Journal of the American Statistical Association*, 106(496), 1394–1404. <https://doi.org/10.1198/jasa.2011.tm10322>
- Davidov, O., & Peddada, S. (2013a). The linear stochastic order and directed inference for multivariate ordered distributions. *The Annals of Statistics*, 41(1), 1–40. <https://doi.org/10.1214/12-AOS1062>
- Davidov, O., & Peddada, S. (2013b). Testing for the multivariate stochastic order among ordered experimental groups with application to dose–response studies. *Biometrics*, 69, 982–990. <https://doi.org/10.1111/biom.12070>
- Demirci, A., & Kartal, M. (2018). Sociocultural risk factors for developmental delay in children aged 3–60 months: A nested case-control study. *European Journal of Pediatrics*, 177(5), 691–697. <https://doi.org/10.1007/s00431-018-3109-y>
- Dubow, E. F., Boxer, P., & Huesmann, L. R. (2009). Long-term effects of parents' education on children's educational and occupational success: Mediation by family interactions, child aggression, and teenage aspirations. *Merrill-Palmer Quarterly*, 55(3), 224–249. <https://doi.org/10.1353/mpq.0.0030>
- Edwards, R. D., & Roff, J. (2010). Negative effects of paternal age on children's neurocognitive outcomes can be explained by maternal education and number of siblings. *PloS One*, 5(9), Article e12157. <https://doi.org/10.1371/journal.pone.0012157>
- Eika, L., Mogstad, M., & Zafar, B. (2019). Educational assortative mating and household income inequality. *Journal of Political Economy*, 127(6), 2795–2835. <https://doi.org/10.1086/702018>
- Flensburg-Madsen, T., & Mortensen, E. L. (2017). Predictors of motor developmental milestones during the first year of life. *European Journal of Pediatrics*, 176(1), 109–119. <https://doi.org/10.1007/s00431-016-2817-4>
- Flouri, E., & Buchanan, A. (2004). Early father's and mother's involvement and child's later educational outcomes. *British Journal of Educational Psychology*, 74(2), 141–153. <https://doi.org/10.1348/000709904773839806>
- Fuller, S., & Hirsh, C. E. (2018). Family-friendly" jobs and motherhood pay penalties: The impact of flexible work arrangements across the educational spectrum. *Work and Occupations*, 46(1), 3–44. <https://doi.org/10.1177/0730888418771116>
- Gollenberg, A. L., Lynch, C. D., Jackson, L. W., McGuinness, B. M., & Msall, M. E. (2010). Concurrent validity of the parent-completed Ages and Stages Questionnaires, 2nd Ed. with the Bayley Scales of Infant Development II in a low-risk sample. *Child Care Health Dev*, 36(4), 485–490. <https://doi.org/10.1111/j.1365-2214.2009.01041.x>
- Guryan, J., Hurst, E., & Kearney, M. (2008). Parental education and parental time with children. *Journal of Economic Perspectives*, 22(3), 23–46. <https://doi.org/10.1257/jep.22.3.23>
- Hale, L., Berger, L. M., LeBourgeois, M. K., & Brooks-Gunn, J. (2009). Social and demographic predictors of preschoolers' bedtime routines. *Journal of Developmental and Behavioral Pediatrics*, 30, 394–402. <https://doi.org/10.1097/DBP.0b013e3181ba0e64>
- Harding, J. F., Morris, P. A., & Hughes, D. (2015). The relationship between maternal education and children's academic outcomes: A theoretical framework. *Journal of Marriage and Family*, 77(1), 60–76. <https://doi.org/10.1111/jomf.12156>
- Hochstedler, K. A., Bell, G., Park, H., Ghassabian, A., Bell, E. M., Sundaram, R., Grantz, K. L., & Yeung, E. H. (2021). Gestational age at birth and risk of developmental delay: The Upstate KIDS Study. *American Journal of Perinatology*, 38(10), 1088–1095. <https://doi.org/10.1055/s-0040-1702937>
- Hoyos Cillero, I., & Jago, R. (2010). Systematic review of correlates of screen-viewing among young children. *Preventive Medicine*, 51(1), 3–10. <https://doi.org/10.1016/j.ypmed.2010.04.012>
- Jeong, J., Kim, R., & Subramanian, S. V. (2018). How consistent are associations between maternal and paternal education and child growth and development outcomes across 39 low-income and middle-income countries? *Journal of Epidemiology and Community Health*, 72(5), 434–441. <https://doi.org/10.1136/jech-2017-210102>
- Jeong, J., McCoy, D. C., & Fink, G. (2017). Pathways between paternal and maternal education, caregivers' support for learning, and early child development in 44 low- and middle-income countries. *Early Childhood Research Quarterly*, 41, 136–148. <https://doi.org/10.1016/j.ecresq.2017.07.001>
- Kalil, A., Ryan, R., & Corey, M. (2012). Diverging destinies: Maternal education and the developmental gradient in time with children. *Demography*, 49(4), 1361–1383. <https://doi.org/10.1007/s13524-012-0129-5>
- Kim, J. (2020). Workplace flexibility and parent-child interactions among working parents in the U.S. *Social Indicators Research*, 151, 427–469. <https://doi.org/10.1007/s11205-018-2032-y>
- Leidy, M. S. (2013). Fathers' contributions to children's social development. *Handbook of Father Involvement* (pp. 158–174). Routledge.
- Li, Z., Kim, R., Vollmer, S., & Subramanian, S. V. (2020). Factors associated with child stunting, wasting, and underweight in 35 low- and middle-income countries. *JAMA Network Open*, 3(4), Article e203386. <https://doi.org/10.1001/jamanetworkopen.2020.3386>
- Lundborg, P., Nilsson, A., & Rooth, D.-O. (2014). Parental education and offspring outcomes: Evidence from the Swedish compulsory school reform. *American Economic Journal: Applied Economics*, 6(1), 253–278. <https://doi.org/10.1257/app.6.1.253>
- Magnuson, K. A., Sexton, H. R., Davis-Kean, P. E., & Huston, A. C. (2009). Increases in maternal education and young children's language skills. *Merrill-Palmer Quarterly*, 55(3), 319–350. <https://doi.org/10.1353/mpq.0.0024>
- McDonald, L., Wardle, J., Llewellyn, C. H., van Jaarsveld, C. H., & Fisher, A. (2014). Predictors of shorter sleep in early childhood. *Sleeping Medicine*, 15(5), 536–540. <https://doi.org/10.1016/j.sleep.2014.01.005>
- McKean-Cowdin, R., Preston-Martin, S., Pogoda, J. M., Mueller, B. A., & Holly, E. A. (2000). Reliability of demographic, smoking and occupational data provided by mothers vs. fathers in a childhood cancer study. *Paediatric and Perinatal Epidemiology*, 14(3), 257–262. <https://doi.org/10.1046/j.1365-3016.2000.00277.x>
- Miclea, D., Peca, L., Cuzmici, Z., & Pop, I. V. (2015). Genetic testing in patients with global developmental delay / intellectual disabilities. A review. *Clujul Medical*, 88(3), 288–292. <https://doi.org/10.15386/cjmed-461>
- Nan, C., Piek, J., Warner, C., Mellers, D., Krone, R. E., Barrett, T., & Zeegers, M. P. (2013). Trajectories and predictors of developmental skills in healthy twins up to 24 months of age. *Infant Behavior and Development*, 36(4), 670–678. <https://doi.org/10.1016/j.infbeh.2013.07.003>
- Ozkan, M., Senel, S., Arslan, E. A., & Karaçan, C. D. (2012). The socioeconomic and biological risk factors for developmental delay in early childhood. *European Journal of Pediatrics*, 171(12), 1815–1821. <https://doi.org/10.1007/s00431-012-1826-1>
- Pakaluk, C. R., & Price, J. (2020). Are mothers and fathers interchangeable caregivers? *Marriage Family Review*, 56(8), 784–793. <https://doi.org/10.1080/01494929.2020.1778318>
- Pancsofar, N., Vernon-Feagans, L., & The Family Life Project Investigators. (2010). Fathers' early contributions to children's language development in families from low-income rural communities. *Early Child Research Quarterly*, 25(4), 450–463. <https://doi.org/10.1016/j.ecresq.2010.02.001>
- Peddada, S. D., Dinse, G. E., & Haseman, J. K. (2005). A survival-adjusted quantal response test for comparing tumour incidence rates. *Journal of the Royal Statistical Society Series C (Applied Statistics)*, 54(1), 51–61. <https://www.jstor.org/stable/3592598>
- Pew Research Center. (2013). *Modern parenthood: Roles of moms and dads converge as they balance work and family*. Pew Research Center.
- Pew Research Center. (2015). *Parenting in America: Outlook, worries, aspirations are strongly linked to financial situation*. Pew Research Center. (https://www.pewresearch.org/social-trends/wp-content/uploads/sites/3/2015/12/2015-12-17_parenting-in-america_FINAL.pdf).

- Pew Research Center. (2023). *Parenting in America today*. Pew Research Center.
- Pougnnet, E., Serbin, L. A., Stack, D. M., & Schwartzman, A. E. (2011). Fathers' influence on children's cognitive and behavioural functioning: A longitudinal study of Canadian families. *Canadian Journal of Behavioural Science*, 43(3), 173–182. <https://doi.org/10.1037/a0023948>
- Rammohan, A., Awofeso, N., & Fernandez, R. C. (2012). Paternal education status significantly influences infants' measles vaccination uptake, independent of maternal education status. *BMC Public Health*, 12, 336. <https://doi.org/10.1186/1471-2458-12-336>
- Rauch, A., Hoyer, J., Guth, S., Zweier, C., Kraus, C., Becker, C., Zenker, M., Huffmeier, U., Thiel, C., Ruschendorf, F., Nurnberg, P., Reis, A., & Trautmann, U. (2006). Diagnostic yield of various genetic approaches in patients with unexplained developmental delay or mental retardation. *American Journal of Medical Genetics Part A*, 140(19), 2063–2074. <https://doi.org/10.1002/ajmg.a.31416>
- Rifas-Shiman, S. L., Rich-Edwards, J. W., Kleinman, K. P., Oken, E., & Gillman, M. W. (2009). Dietary quality during pregnancy varies by maternal characteristics in Project Viva: a US cohort. *Journal of the American Dietetic Association*, 109(6), 1004–1011. <https://doi.org/10.1016/j.jada.2009.03.001>
- Roberts, E., Bornstein, M. H., Slater, A. M., & Barrett, J. (1999). Early cognitive development and parental education. *Infant and Child Development*, 8, 49–62. [https://doi.org/10.1002/\(SICI\)1522-7219\(199903\)8:1<49::AID-ICD188>3.0.CO;2-1](https://doi.org/10.1002/(SICI)1522-7219(199903)8:1<49::AID-ICD188>3.0.CO;2-1)
- Rowe, M. L. (2012). A longitudinal investigation of the role of quantity and quality of child-directed speech in vocabulary development. *Child Development*, 83(5), 1762–1774. <https://doi.org/10.1111/j.1467-8624.2012.01805.x>
- Rubin, D. B. (1987). *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons.
- Slykerman, R. F., Thompson, J. M. D., Clark, P. M., Becroft, D. M. O., Robinson, E., Pryor, J. E., Wild, C. J., & Mitchell, E. A. (2007). Determinants of developmental delay in infants aged 12 months. *Paediatric and Perinatal Epidemiology*, 21, 121–128. <https://doi.org/10.1111/j.1365-3016.2007.00796.x>
- Sonnander, K., & Claesson, M. (1999). Predictors of developmental delay at 18 months and later school achievement problems. *Developmental Medicine and Child Neurology*, 41, 195–202. <https://doi.org/10.1111/j.1469-8749.1999.tb00579.x>
- Squires, J., & Bricker, D. (2009). *Ages & Stages Questionnaires®, Third Edition (ASQ®-3): A Parent-Completed Child Monitoring System*. Paul H. Brookes Publishing Co., Inc.
- Staples, A. D., & Bates, J. E. (2018). Parenting of infants and toddlers. In In. M. R. Sanders, & A. Morawska (Eds.), *Handbook of Parenting and Child Development Across the Lifespan* (pp. 585–607). Springer International Publishing. https://doi.org/10.1007/978-3-319-94598-9_26
- Tamis-LeMonda, C. S., Bornstein, M. H., Baumwell, L., & Damast, A. M. (1996). Responsive parenting in the second year: Specific influences on children's language and play. *Early Development and Parenting*, 5(4), 173–183. [https://doi.org/10.1002/\(SICI\)1099-0917\(199612\)5:4<173::AID-EDP131>3.0.CO;2-V](https://doi.org/10.1002/(SICI)1099-0917(199612)5:4<173::AID-EDP131>3.0.CO;2-V)
- Telias, A., Narea, M., & Abufhele, A. (2022). A mediation analysis to disentangle relations between maternal education and early child development. *International Journal of Behavioral Development*, 46(6), 568–575. <https://doi.org/10.1177/01650254221108143>
- Trinh, M. H., Sundaram, R., Robinson, S. L., Lin, T.-C., Bell, E. M., Ghassabian, A., & Yeung, E. H. (2020). Association of trajectory and covariates of children's screen media time. *JAMA Pediatrics*, 174(1), 71–78. <https://doi.org/10.1001/jamapediatrics.2019.4488>
- Valla, L., Wentzel-Larsen, T., Hofoss, D., & Slinning, K. (2015). Prevalence of suspected developmental delays in early infancy: Results from a regional population-based longitudinal study. *BMC Pediatrics*, 15, 215. <https://doi.org/10.1186/s12887-015-0528-z>
- van Buuren, S., & Groothuis-Oudshoorn, K. (2011). mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software*, 45(3), 1–67. <https://doi.org/10.18637/jss.v045.i03>
- Vollmer, S., Bommer, C., Krishna, A., Harttgen, K., & Subramanian, S. V. (2017). The association of parental education with childhood undernutrition in low- and middle-income countries: Comparing the role of paternal and maternal education. *International Journal of Epidemiology*, 46(1), 312–323. <https://doi.org/10.1093/ije/dyw133>
- Yeung, E. H., Sundaram, R., Bell, E. M., Druschel, C., Kus, C., Ghassabian, A., ... Buck Louis, G. M. (2016). Examining infertility treatment and early childhood development in the Upstate KIDS study. *JAMA Pediatrics*, 170(3), 251–258. <https://doi.org/10.1001/jamapediatrics.2015.4164>
- Yue, A., Jiang, Q., Wang, B., Abbey, C., Medina, A., Shi, Y., & Rozelle, S. (2019). Concurrent validity of the Ages and Stages Questionnaire and the Bayley Scales of Infant Development III in China. *PLoS One*, 14(9), Article e0221675. <https://doi.org/10.1371/journal.pone.0221675>
- Zablotsky, B., Black, L. I., Maenner, M. J., Schieve, L. A., Danielson, M. L., Bitsko, R. H., Blumberg, S. J., Kogan, M. D., & Boyle, C. A. (2019). Prevalence and trends of developmental disabilities among children in the United States: 2009–2017. *Pediatrics*, 144(4), Article e20190811. <https://doi.org/10.1542/peds.2019-0811>
- Zadeh, Z. Y., Farnia, F., & Ungerleider, C. (2010). How home enrichment mediates the relationship between maternal education and children's achievement in reading and math. *Early Education Development*, 21(4), 568–594. <https://doi.org/10.1080/10409280903118424>